SELVAGANAPATHY K

AI/ML Engineer | Computer Science Student

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CAREER OBJECTIVE

AI/ML Engineer and Computer Science student with a strong foundation in Java, Python, Data Structures & Algorithms, Machine Learning, Deep Learning, and Computer Vision. Experienced in building and deploying end-to-end AI applications, from model training to web-based deployment. Seeking an opportunity to apply and expand technical skills, gain hands-on industry experience, and contribute meaningfully to an organization's engineering goals.

EDUCATION

Bachelor of Engineering – Computer Science and Engineering

Bannari Amman Institute of Technology | Expected 2028 | CGPA: 7.9 / 10.0

TECHNICAL SKILLS

Languages: Python, Java, C, HTML, CSS, JavaScript

AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, YOLOv8, TensorFlow, OpenCV, Scikit-learn, NumPy, Pandas

Tools: Git, GitHub, VS Code, Streamlit, Jupyter Notebook

Currently Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), MLOps, Advanced Java DSA

Core Subjects: DSA, OOP, DBMS, Operating Systems

PROJECTS

Computer Vision-Based Automatic Waste Classification System

Python, YOLOv8, OpenCV, Deep Learning, Streamlit

- Built an AI-powered waste classification system using YOLOv8 to classify waste into Organic, Recyclable, and Hazardous categories via Computer Vision.
- Trained a custom object detection model and developed a modern web interface for real-time predictions.
- Designed and deployed the complete pipeline end-to-end, from model training to a live web application with a REST API backend.

GitHub | Live Demo | API Docs

Face Emotion Recognition

Python, TensorFlow, OpenCV, Streamlit

- Built a deep learning model to detect facial emotions — Happy, Sad, Angry, Fear, Surprise, Disgust, and Neutral — from images and webcam input.
- Developed an interactive Streamlit web application for real-time emotion prediction.

GitHub | Live Demo

Employee Attrition Prediction

Python, Scikit-learn, Pandas, NumPy, Streamlit

- Developed a Machine Learning application that predicts employee attrition using HR analytics data.
- Built an interactive interface that analyzes employee information and predicts the likelihood of an employee leaving the organization.

GitHub | Live Demo

CERTIFICATIONS

- **Oracle Certified Foundations Associate – Java SE 17** — Core Java, OOP, Collections, Exception Handling

ACHIEVEMENTS

- Oracle Certified Foundations Associate – Java SE 17.
- Participated in AI and innovation hackathons.
- Developed multiple AI and Computer Vision projects, including a YOLOv8-based waste classification system.
- Built and deployed Machine Learning applications using Streamlit, with active GitHub contributions.

STRENGTHS & LANGUAGES

Strengths: Problem Solving, Teamwork, Quick Learner, Communication, Analytical Thinking

Languages: Tamil, English